IBM HR ANALYTICS EMPLOYEE ATTRITION

SQL Analysis Report

Prepared by: **Sarthak Sehgal**

Executive Summary

This project analyzes employee attrition in the IBM HR dataset to identify workforce segments associated with higher employee turnover. The analysis focuses on department, overtime, job satisfaction, and monthly income. SQL queries are used to inspect data quality, validate key fields, calculate attrition rates, and produce grouped results that can support HR decision-making.

The original report identifies higher attrition in Sales, among employees working overtime, among employees with lower job satisfaction, and within lower monthly-income bands. These relationships highlight areas for further investigation and potential employee-retention initiatives.

Analysis areas: Department • Overtime • Job Satisfaction • Monthly Income

Dataset details and reported findings are retained from the original report. SQL statements are provided as a reproducible analysis framework and have not been executed against the user's database.

Business Objective

The objective is to understand which employee groups show higher attrition and provide evidence to help the organization investigate retention risks.

Key business questions

• Which department has the highest employee attrition rate?

• How does attrition differ between employees who work overtime and those who do not?

• Is lower job satisfaction associated with higher attrition?

• How does attrition vary across monthly-income bands?

Dataset Overview

Dataset attribute	Description
Dataset	IBM HR Employee Attrition
Employee records	1,470
Original columns	35
Columns after reported cleaning	32
Target variable	Attrition (Yes / No)

Data Quality and Preparation

The original report documents the following checks and preparation decisions:

• Missing values: none reported; no imputation was required.

• Duplicate records: none reported.

• Employee identifiers: EmployeeNumber values were unique.

• Constant columns: EmployeeCount, Over18, and StandardHours were removed because they did not vary.

• Data types and tenure-related fields: reviewed for consistency.

• Outliers: reviewed and retained where they represented valid employee records.

Dataset preview

```
SELECT *
FROM employee_attrition
LIMIT 10;
```

Record count

```
SELECT COUNT(*) AS total_rows
FROM employee_attrition;
```

Missing-value checks

```
SELECT
  COUNTIF(Department IS NULL) AS missing_department,
  COUNTIF(Attrition IS NULL) AS missing_attrition,
  COUNTIF(OverTime IS NULL) AS missing_overtime,
  COUNTIF(JobSatisfaction IS NULL) AS missing_job_satisfaction,
  COUNTIF(MonthlyIncome IS NULL) AS missing_monthly_income
FROM employee_attrition;
```

COUNTIF is BigQuery syntax. For other SQL platforms, use SUM(CASE WHEN column_name IS NULL THEN 1 ELSE 0 END).

Duplicate employee identifiers

```
SELECT EmployeeNumber, COUNT(*) AS record_count
FROM employee_attrition
```

```
GROUP BY EmployeeNumber
HAVING COUNT(*) > 1;
```

Review constant columns

```
SELECT
  COUNT(DISTINCT EmployeeCount) AS employee_count_values,
  COUNT(DISTINCT Over18) AS over18_values,
  COUNT(DISTINCT StandardHours) AS standard_hours_values
FROM employee_attrition;
```

Validate key categories

```
SELECT Attrition, COUNT(*) AS employees
FROM employee_attrition
GROUP BY Attrition;

SELECT OverTime, COUNT(*) AS employees
FROM employee_attrition
GROUP BY OverTime;

SELECT JobSatisfaction, COUNT(*) AS employees
FROM employee_attrition
GROUP BY JobSatisfaction
ORDER BY JobSatisfaction;
```

Analysis and Findings

Department Attrition

SQL objective: compare headcount and attrition rate by department.

```
SELECT
  Department,
  COUNT(*) AS total_employees,
  SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS employees_left,
  ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END)
        / NULLIF(COUNT(*), 0), 2) AS attrition_rate_pct
FROM employee_attrition
GROUP BY Department
ORDER BY attrition_rate_pct DESC;
```

Reported finding: Sales has the highest attrition rate at 21%; HR is 19%; Research & Development is lowest at 14%.

Overtime and Attrition

```
SELECT
  OverTime,
  COUNT(*) AS total_employees,
  SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS employees_left,
  ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END)
        / NULLIF(COUNT(*), 0), 2) AS attrition_rate_pct
FROM employee_attrition
GROUP BY OverTime
ORDER BY attrition_rate_pct DESC;
```

Reported finding: employees working overtime have a 31% attrition rate, compared with 10% for employees who do not work overtime.

Job Satisfaction and Attrition

```
SELECT
  JobSatisfaction,
  COUNT(*) AS total_employees,
  SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS employees_left,
  ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END)
        / NULLIF(COUNT(*), 0), 2) AS attrition_rate_pct
FROM employee_attrition
GROUP BY JobSatisfaction
ORDER BY JobSatisfaction;
```

Reported finding: job-satisfaction level 1 has the highest attrition rate at 23%; level 4 has the lowest at 11%. Levels 2 and 3 are reported at 16% and 17%, respectively.

Monthly Income and Attrition

```
SELECT
  CASE
    WHEN MonthlyIncome BETWEEN 1009 AND 6008 THEN '1009-6008'
```

```
    WHEN MonthlyIncome BETWEEN 6009 AND 11008 THEN '6009-11008'
    WHEN MonthlyIncome BETWEEN 11009 AND 16008 THEN '11009-16008'
    WHEN MonthlyIncome BETWEEN 16009 AND 21008 THEN '16009-21008'
    ELSE 'Outside reported bands'
  END AS income_band,
  COUNT(*) AS total_employees,
  SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END) AS employees_left,
  ROUND(100.0 * SUM(CASE WHEN Attrition = 'Yes' THEN 1 ELSE 0 END)
        / NULLIF(COUNT(*), 0), 2) AS attrition_rate_pct
FROM employee_attrition
GROUP BY income_band
ORDER BY income_band;
```

Reported finding: the 1,009–6,008 income band has the highest attrition rate at 20%; the 16,009–21,008 band has the lowest at 4%. Verify band boundaries and sort order against the actual dataset before publication.

Consolidated Findings

• Sales has the highest reported department attrition rate (21%).

• Employees working overtime have a higher reported attrition rate (31%) than those not working overtime (10%).

• Employees with job-satisfaction level 1 show the highest reported attrition rate (23%).

• The lowest reported monthly-income band has a 20% attrition rate, compared with 4% in the highest band.

These are associations observed in the report and should not be interpreted as proof of causation.

Recommendations

• **Strengthen Sales retention:** review workload, management support, recognition, and career-development opportunities.

• **Review overtime practices:** assess staffing, workload distribution, and work-life balance.

• **Improve job satisfaction:** use employee feedback, engagement initiatives, recognition, and progression pathways.

• **Review compensation:** assess salary progression and growth opportunities for lower-income employee groups.

Conclusion

The SQL analysis framework provides a structured way to measure attrition across key workforce segments. The original report points to overtime, job satisfaction, monthly income, and department as priority areas for further review. HR teams should validate the patterns with employee feedback and additional analysis before deciding on retention interventions.

SQL Execution Notes

• Replace employee_attrition with the actual table name in your database.

• The COUNTIF query uses BigQuery syntax; adjust it if using MySQL, PostgreSQL, SQL Server, or another database.

• Run the queries on the imported dataset and verify output percentages before publishing.

Accuracy note: the percentages shown in this report come from the original report; they have not been independently recalculated by executing these SQL queries against the source dataset.